

## EnviroTime: Digital Clock with Real-Time Temperature Monitoring

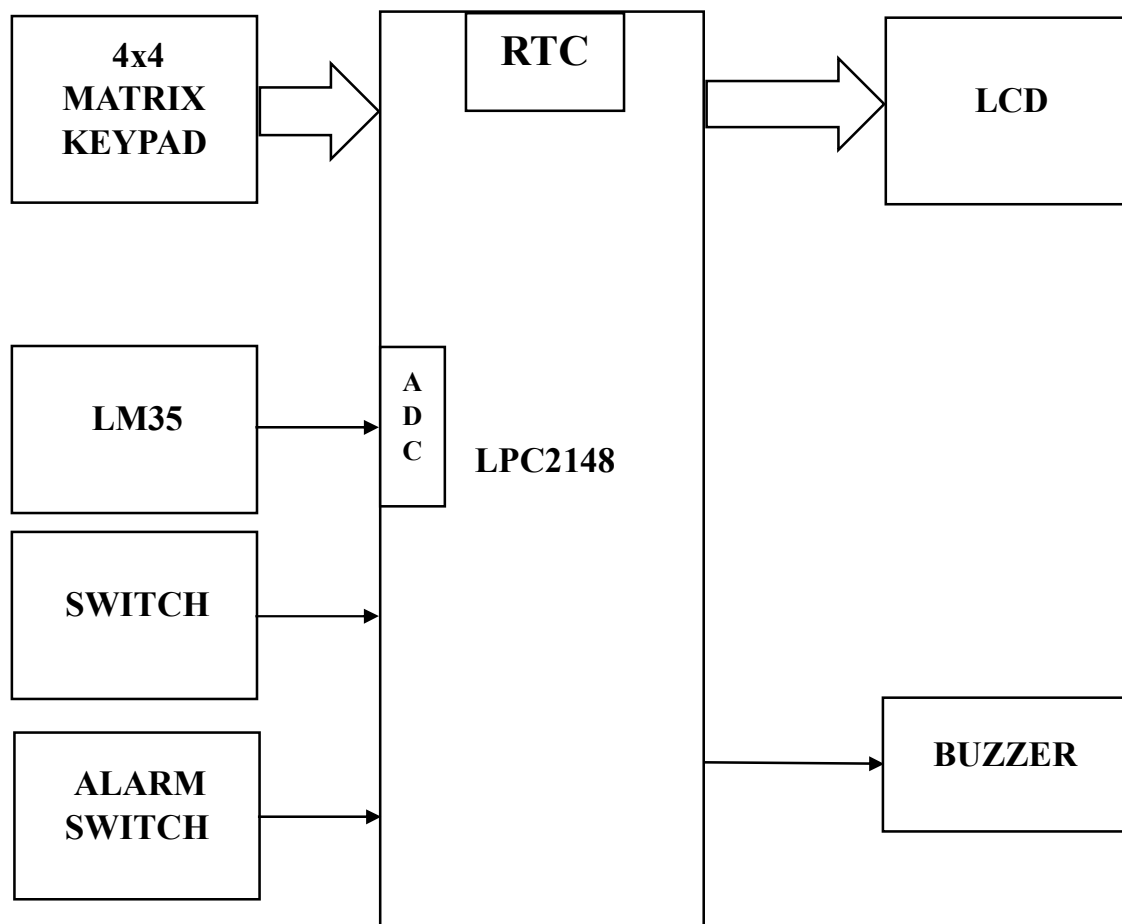
### AIM:

To design and develop a real-time embedded system using the LPC2148 microcontroller that displays accurate digital clock information along with continuous ambient temperature monitoring, while allowing secure user-controlled modification of time and alarm settings through switch-based polling and keypad authentication.

### Objectives:

- Implement Real-Time Clock (RTC) for accurate display of time, date, day, month, and year
- Monitor ambient temperature using LM35 sensor through on-chip ADC
- Display real-time clock and temperature information on a 16×2 LCD
- Provide alarm setting functionality for user-defined time alerts
- Interface keypad for secure password-based user input and parameter modification
- Use switch-based polling to enter edit mode for updating RTC and alarm settings
- Generate buzzer alerts for alarm indication and system notifications

### BLOCK DIAGRAM:



## **REQUIREMENTS:**

### **HARDWARE REQUIREMENTS:**

- LPC2148
- 16X2 LCD
- 4X4 MATRIX KEYPAD
- SWITCH
- LM35
- BUZZER
- USB-UART CONVERTER / DB-9 CABLE

### **SOFTWARE REQUIREMENTS:**

- EMBEDDED-C PROGRAMMING
- FLASH MAGIC

### **PROJECT WORKFLOW:**

EnviroTime is a real-time embedded system designed using the LPC2148 microcontroller to display accurate digital clock information along with continuous ambient temperature monitoring.

The system uses a Real-Time Clock (RTC) module to maintain and display current time, date, day, month, and year, while a temperature sensor such as LM35 is interfaced through the on-chip ADC to measure and display live room temperature. The information is continuously updated on a 16×2 LCD, providing a compact and user-friendly monitoring system.

The system also provides secure user-controlled settings through keypad input and switch-based access for modifying time parameters and alarm settings.

This project is highly useful in homes, offices, laboratories, hospitals, and industrial environments where both timekeeping and environmental monitoring are important.

### **System Working:**

Initialize all required peripherals and start the application logic. The required peripherals include:

- RTC
- LCD
- Switch input
- Keypad for user input
- ADC for temperature monitoring
- Buzzer for alarm/security indication

After initialization, the system enters continuous monitoring mode.

### **Continuous Monitoring Operation**

Inside the infinite loop, the controller performs the following operations continuously:

#### **Read and Display RTC & ADC Information on LCD**

Read the current RTC parameters: Hour, Minute, Second, Day, Date, Month and Year

Display the RTC information on the LCD in formatted form.

Read the analog temperature sensor value through ADC, convert it into temperature, and display it on the LCD along with system status.

Example display: HH:MM:SS    T:28°C

DD/MM/YYYY DAY

When the alarm time is reached, the system generates an indication through the buzzer. To stop the alarm tone, the user must press the alarm switch, after which the buzzer will be turned OFF.

### **Switch Polling for User Editing:**

When the user presses the edit switch, the controller detects it by continuously checking the switch status in the main loop. Once the switch press is detected, the system enters the editing process.

### **Basic Level Security Inclusion**

Before allowing any modification, the system asks for a basic password.

### **Security Check Before Editing**

Once the switch press is detected:

- Prompt user to enter password through keypad
- Compare entered password with stored password
- If password is correct:
  - allow editing of parameters
- If password is wrong:
  - deny access
  - increment wrong-attempt counter
  - show “Access Denied” on LCD
  - optionally blink buzzer/LED

### **Wrong Attempt Handling**

- If wrong password is entered 3 times:
  - lock edit mode for a short duration or until reset
  - show “System Locked”
  - optional alert indication through buzzer

This gives the project a simple but attractive security layer.

### **Editable Parameters in Secure Edit Mode**

After successful password validation, allow user to edit:

- Current RTC time, day and date
- Alarm time settings
- Password change option

The keypad can be used to navigate through a menu.

### **Return to Normal Mode**

After editing is completed:

- save updated parameters
- exit edit mode and return to normal monitoring operation

The system then resumes:

- RTC display
- temperature monitoring
- alarm monitoring and buzzer indication

\*\*\*\*\* ALL THE BEST \*\*\*\*\*

### **Note:**

**The application code must be developed following industry-standard embedded programming practices, with proper structure, naming conventions, modular design, and complete test case validation support. Any code not adhering to these standards will not be accepted.**